Njord Challenge 2026 Technical Report Humber ASV – Loon-E

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***Abstract*—The Loon-E is Humber ASV's first Autonomous Surface Vehicle (ASV), developed for the 2026 RoboBoat competition and further improved to participate in the Njord Challenge. This report presents the rationale behind design decisions which contributed to the development of the Loon-E, including the team's objectives, overall system logic, and a proposed methodology for the execution of the competition's various tasks. The Vessel Design and Software sections include a description of each subsystem and the process of their development, including the hardware, electrical, and software design. Finally, the various tests performed to verify the functionality of each subsystem are discussed, whether in simulation or in real life.**

Keywords—Artificial intelligence, Autonomous vehicles, Design Methodology, Marine engineering

I. Introduction

This will be Humber ASV's first year participating in the Njord Challenge. The team's objective was to design and build a functional remote-controlled boat and add any systems necessary for full autonomy. Autonomy requires the integration of sensors (camera and GPS) as well as the software which integrates these sensors into the overall control flow (see Figure 1 and Appendix A).

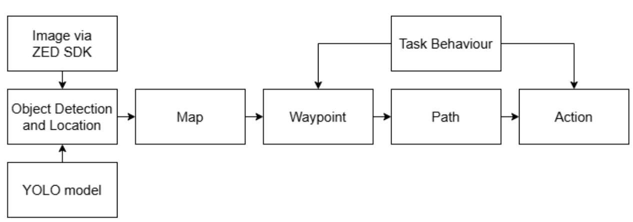

Figure 1: Control Flow

II. System Overview

As a first attempt at an ASV, there will inevitably need to be adjustments made if this project should continue for future years. As such, iterability and adaptability is a large focus throughout development. For this first iteration, components were selected based on preliminary research with particular focus on the equipment used by other ASVs used in competitions such as RoboBoat and the Njord Challenge. Some custom elements were included to allow students to be more involved in the design process; these are discussed in the later sections.

A. Hull Design

The Loon-E was designed as a catamaran to allow for increased stability and a large area in which to place any necessary components. The hull was intentionally designed with parting lines and draft directions and angles required for traditional molding techniques such as to be manufacturable using a variety of methods, such as fiberglassing, roto-molding, and 3D printing.

The hull notably features panels on the front and back of the boat on which various modules may be installed, such as the firebox in which the batteries will be located. This makes it so that if a new element is required to expand the capacities of the Loon-E, they can be easily added to the hull with minimal changes to design.

After an initial fiberglass test, it proved too difficult to pursue this methodology for manufacturing the hull and the team opted instead for 3D printing. To achieve this, the hull was segmented into parts, each of which was printed individually with ABS on a smaller printer. The pieces were then glued, assembled, and sealed using melted ABS, after which it is fully waterproof. Once set, the hull could finally be sanded, primed, and painted.

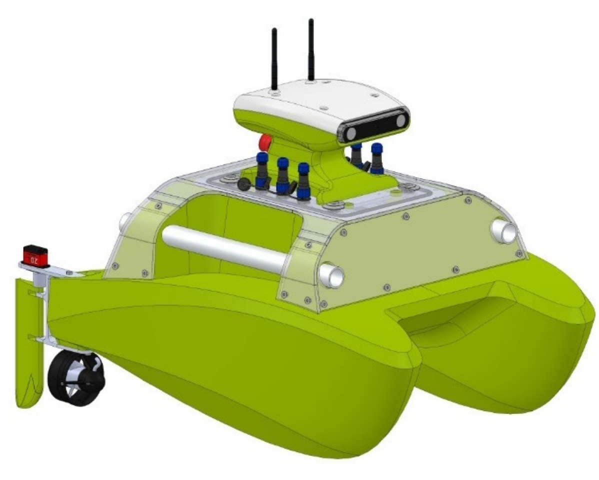

Fig. 2. Loon-E CAD

The hull was initially proposed to be printed in a single piece on a sufficiently large 3D printer. To accommodate the printer selected, the hull was sized at approximately 60 cm in width by 1 m in length (See Appendix C). The full height of the boat with all components included is 75 cm tall, featuring a separately printed head in which sensors are installed.

B. Propulsion

The size of the boat is on the smaller end due to 3D printing constraints and to allow for it to travel greater speeds. As such, the propulsion system follows the same philosophy, keeping it parts and weight to a minimum. The Loon-E uses two Blue Robotics T200 thrusters mounted to the back of the boat, driving the boat through differential thrust. The propellers are powered at 20V to utilize their maximum thrust. At the maximum allowable setting for them, each propeller uses 551 W, producing a joint thrust of about 12.5 kg*f.

For more precise steering, rudders have also been installed behind each propeller. The rudder profile was selected based on available research on the topic and complemented by creating a test environment many different profiles, and lengths (see Section V). Each rudder is individually controlled by a servo motor, but control of both rudders is done through a single signal to simplify the range of movements of the boat. Components are listed in Appendix B.

C. Batteries and Electrical System

A full electrical system was implemented for the first time this year. The system has two different power sources: Two 20 V, 6 Ah lithium-ion batteries, which will only be used to power the propellers, and a 14.8 V, 18 Ah battery which will be used to power the rest of the system, as well as the propellers in case of emergency.

The reasoning behind the battery selection was to extract all the possible power out of the propellers used, as well as offering redundancy for the system, increasing the maximum run time of the boat. For this last point, a switching system has been added enabling the 14.8 V battery to be used.

The batteries are of significant volume and weight, reaching an accumulated weight of 2.95 kg. For this reason, they were placed near the bow of the boat, inside the firebox, to bring the center of mass closer to the middle as the electronics in the inside and the rest of the components are located nearer to the stern of the boat.

The electrical team also developed custom PCBs. The first is used to alternate between the RC and Autonomous modes. It utilizes 2:1 multiplexer ICs that select the signals coming from the Jetson or from the RC receiver. It also sends a signal back to the Jetson indicating which mode the boat is being driven in. The second PCD developed is a light indicator located on the to head of the ship. It mainly consists of an RGB cob and a driver IC. The colour of each panel can be controlled though individual PWM signals raging from 0.3 V to 2.5 V. The last PCB developed was a voltage sensor. It consists of a voltage divider and an Analog to Digital Converter that relays the data to the CPU using the I2C protocol. The design of all PCBs was done using the EasyEDA software and diagrams for these circuits can be found in Appendix D.

D. Sensors

The most crucial information for the Njord challenge includes a method to identify the ASV's position as well as the position of the objects around it. To acquire the boat's position a GPS is used. From it, additional information such as speed and heading can be derived.

Initially the team intended on utilizing a MT3337 Chip Set GPS module, as well as an Adrafruit BNO055 9-axis IMU. Thorough calibration, filtering and sensor fusion was attempted to combine these sensors into a reliable location, heading and speed readings. However, given the capabilities of the GPS sensor (see Section V) and the limited budget, the team decided to utilize the sensors and sensor fusion capabilities of a smartphone.

The system implemented uses an Android smartphone purely as a data acquisition platform. The phone acquires heading, speed, latitude, and longitude information using its internal sensors and transmits the data over a wired USB connection. It does not require nor uses Wi-Fi or any other form of wireless communication. The Android application was developed in Android Studio using Kotlin, and the receiving side was implemented in Python. Communication between the phone and the computer is performed through ADB (Android Debug Bridge) reverse TCP tunneling over USB.

The heading value is obtained using Android's "TYPE_ROTATION_VECTOR" sensor fusion system, utilizing the sensor data coming from the "SensorManager" system service. This is a high-level orientation estimate generated internally by Android using the phone's gyroscope, accelerometer and magnetometer. Latitude and longitude are acquired from the phone's internal GNSS/GPS subsystem through Android's "LocationManager" system service using the GPS provider. Speed is acquired from Android's GPS location system through "location.speed" variable, belonging to the "Location Manager" mentioned earlier. This provides a speed estimate in meters per second, using the Doppler-based GNSS velocity estimation from the GPS chipset itself. Tests to validate this system are further described in Section V.

To detect the environment, the boat needs to acquire image data to identify shapes, data matrices, and objects. It also requires distance data to adequately map and plan its way around. Solutions such as LiDAR, optical cameras and sonars were assessed. The team decided on using a stereoscopic camera such as Stereolabs' ZED X due to its ability to provide distance data between the boat and surrounding objects as well as image data in a single harmonized environment. Data from the camera can be obtained through use of the ZED SDK [2], and up to three additional Stereolabs cameras may be added should it be deemed necessary in future years.

E. Hardware

The Loon-E's central computer is the Jetson Orin Nano Developer Kit. It is also equipped with an Ubiquiti Bullet to allow for it to send data to the base station.

The base station hardware is recommended to uses a quad-core 2.7GHz processor, 16Gb RAM, an Nvidia GPU with compute capabilities >= 7.5, 6 GB of VRAM, Windows 10 or Ubuntu Jammy Jellyfish [x], and internet capabilities – Though not recommended, software on the base station can also be run on MacOs.

The web client used to view the Graphical User Interface (see Section IV) relies on any modern mobile device as it runs entirely on the browser. These are used alongside a switch and a router for internet connection.

F. Safety Systems

The vessel carries the base safety features as required for the Njord Challenge. Firstly, it has an on-board e-stop as well as a remote kill switch, controlled by the RC remote. In case the vessel loses connection with the RC remote, the receiver is configured to turn off the channel corresponding to the remote e-stop, shutting it off and cutting power to the propellers.

Regarding the firebox, all three batteries are located inside a fire-resistant enclosure which carries a UL flammability rating of 94V-0, and an Ingress Protection rating of IP54, but can be upgraded to IP66 with gaskets and screws with O-rings.

To connect devices on the exterior of the hull, waterproof connectors were used. These can be removed and sealed when not in use. To wire devices on the outside, watertight heat shrinking tubing was used to prevent water ingress to the bare connections.

The decision to run the propellers at their full potential was made early on the project; for this reason, it was important to ensure that the system match all other conditions. Robust components that allowed normal operation at 30A or above were implemented, namely terminal blocks, waterproof connectors, relays, as well as utilizing 10 AWG wire in the propulsion system circuitry.

Similarly, to ensure consistent and secure connection, ferrules, WAGO connectors and screw-on terminal blocks were used around the system. The reasoning behind this was to avoid soldering as much as possible: soldering connections are permanent and time consuming and introduce potential points of failure as a proper connection is hard to standardize and difficult to visually confirm. However, soldering was utilized on the exterior of the boat (ex. to connect to motors) as non-permanent connections are harder to waterproof.

Breakers were added to the 20 V batteries tripping at 30 A, slightly above the maximum operating current of 28.26 A. These batteries are monitored by a GoBILDA Battery Management System (BMS) that will stop the battery from discharging if it reaches the minimum safe voltage.

The individual level of each cell on the 14.8 V is also monitored through the BMS. If any of the cells is depleted, the output to the BMS is cut off. A voltage sensor that relays information to the RC remote was placed at the output of the BMS. In this manner, if the lithium-ion battery ever falls under the minimum safe voltage, the voltage will not have a reading. Additional, voltage sensors are placed for all three batteries and readings are continuously sent to the ground station.

III. Software

A. Software Design and Architecture

The overall operation of the system uses ROS 2 architecture, with nodes for each of the system's core processes (outlined in the Implementation section) each contained in a single package. The process for executing each task is also outlined in its own node and is executed when required (ie. the boat will be set to execute the Maneuvering and Path Finding program when attempting it and will be provided the GPS coordinates before the attempt). This architecture allows for a more straightforward representation of each process, supporting future development. Python is used for this project due to the team's familiarity with the language.

B. Task Strategy

Each task is divided into a series of subtasks, such as moving towards a GPS point, stopping, or turning. The active subtask is changed upon some condition being met, such as the boat arriving at a given GPS coordinate, a given object being visible, or the location of that object in the frame. An example is shown below for the Maneuvering and Path Finding task; task strategy is outlined in greater detail in Appendix E.

C. Implementation

The standard operation of the Loon-E outside of initialization and shutdown procedures can be divided into four phases, as outlined below. In addition to this, data is sent to the GUI after every loop – This is described further in Section IV.

1) Object Detection

The Navier dataset was used to train the YOLO11 object detection model [4], with the marker image dataset being used for verification. This includes labels for the six types of buoys (Red, Green, North, East, South, West). In addition to this, images were sourced online of various types of boats, including the Maritime Robotics Otter, and labelled using Roboflow to allow the model to identify boats for the Collision Avoidance task.

2) Mapping

Using the ZED SDK [2] it is possible to measure the position of detected objects in the image relative to the camera and place these in a local map. The map represents what is currently visible through the ZED X camera from a top view, including the classes of detected objects, and each cell represents a square with sides of 0.5 m. Each buoy (dia. 0.35 m) is assumed to take up the entirety of the nearest space on this grid, providing a buffer in case of any rounding errors.

This information can then be combined with information on the ASV's current position and heading to add this data to a global map which represents the full explored area. Positions relative to relevant objects within this map can then be converted to GPS coordinates which are used as waypoints for path planning.

3) Path Planning

Provided a point on the map and a point corresponding to the boat's current position, a straight line is drawn between these two points. If an object is within 1 m of a point on this path, the A* algorithm [5] will be used to adjust the path such that it avoids the obstacle. This allows for faster path planning in scenarios where there are no obstacles to avoid. To allow for a smoother transition between points, it is ensured that points are at least 2 m (two boat lengths) apart. The vector which can be made between these two points can also be calculated at this step and used as the target heading, used in the next step.

4) Following the Path

If the boat is more than 45° away from the target heading, it will stop and turn toward the target heading. If it is within this range, it will drive towards the point indicated during the path planning step.

The Loon-E drives primarily using differential thrust, where the two propellers may rotate at different speeds to allow for turning. When differential thrust is used, the outside propeller (ex. the right propeller when turning left) will be set to the maximum PWM value, and the inside propeller will be set to some value proportional to the difference between the boat's current heading and its target heading. This value is calculated through a PID control algorithm.

Turning can be further supplemented using rudders which can be positioned at one of three positions: -35°, 0°, and 35° - this was selected as 35° is the proposed maximum angle for a rudder [6]. Using this system, the rudders will be active if the difference between the boat's current heading and the target heading is sufficiently large. This methodology reduces the number of possible states of the propulsion system (varying both the thrusters and the rudders fully would be unnecessarily complex at this stage of development).

The boat's current speed is also checked at this stage, decreasing or increasing the power of each propeller by a proportional amount to approach the desired speed. A flowchart outlining this process can be found in Appendix E.

IV. Graphical User Interface

The graphical user interface is hosted on a server using React and MUI. Users use a browser to connect to www.humberasv.ca/connect; it first displays a form to enter a security token. The token is generated through an admin site hosted on the base-station. Once they have entered and clicked submit with a valid token, the browser will attempt to connect directly to a ROS telemetry node which relays the ASV's telemetry and other data to their browser.

There are various elements on the GUI, shown in the figure below. The control panel shows the power applied to the propellers and the rudder angles. A heading is displayed on the top, and below that a task header, with task information, and system checks. The graphical boxes that outline labels and object detected by the ASV are rendered in the base station to reduce operational load. The screen border come in various colors to notify the user what status the ASV is in.

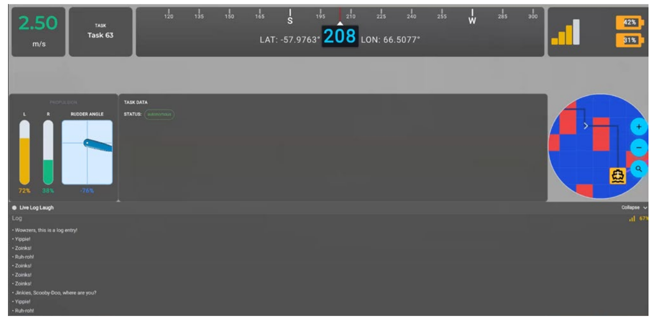

Fig. 3. Functional GUI Testing

V. Innovative Aspect

The electronics hatch is made watertight with latches and a rubber seal that is squeezed when closed. Two electronics plates are mounted to the Loon-E's lid using aluminium brackets, onto which various electronics components may be mounted. The plates may be made in aluminium to serve as an additional heat sink or acrylic for faster prototyping, especially when it comes to placement of mounting holes for the components.

Although in theory this is sufficient to keep the electronics safe, the electronics devices are suspended on the air (see Appendix C, "Electronics Hatch"). This should provide extra protection in case of a any leak maintaining the water at the bottom of the cavity and away from the electronics.

Designed in this way, it was possible to maximize the available space for electrical components and allow for modularity of the electrical system (more easily replacing or adding components) provided there is sufficient room for any new components. Placement of these components can be tested in a CAD software using 3D models and parts may also "overlap" by varying their heights if required.

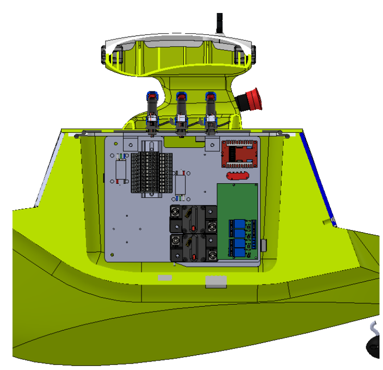

Fig. 4. Loon-E Cross-Section, showing Electronics Hatch

Another innovative feature of the electrical system is a secondary backup system, further described in Section II. By default, the system utilizes two 20 V batteries to control the Loon-E's propellers, the capacity of each battery being 6 Ah. Although this should be enough to perform tests and attempts at the competition, the team did not want it to be a limiting factor. Therefore, it was proposed that the 14.8 V battery that is regularly used to power the electronics should alternatively be used to power the propellers if needed.

As shown on the diagram below (see Figure 5), the driver can alternate between two sets of relays that direct power to the propellers. In this way, all functionality of the remote E-Stop is maintained. Use of the 14.8 V battery to power the propellers is mainly intended in case of emergency so that the Loon-E can be safely returned to shore using RC control, as it is unclear what effect it would have on the full system long term.

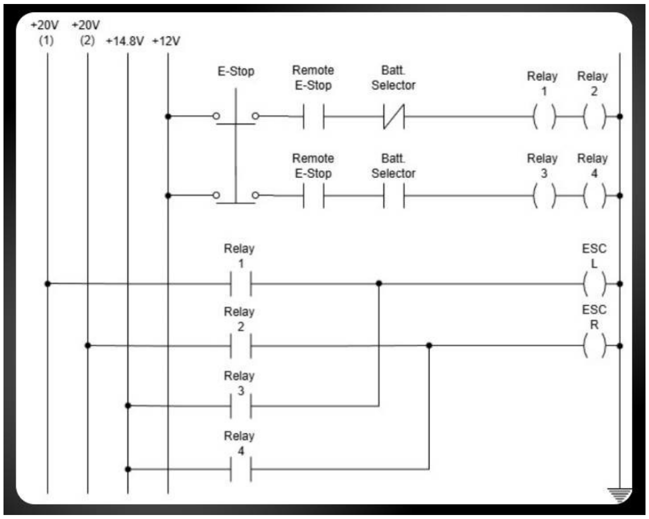

Fig. 5. Electronic Diagram for Dual Battery System

VI. Testing Strategy

Testing was done in various phases, starting with testing of individual sensors or subsections and expanding over time until the full system could be tested. Tests were mainly done with the RoboBoat competition tasks in mind and examples shown may be reflective of this. Tests are described here in order of subsystem as outlined in Sections II and III, rather than being in chronological order.

A. *Rudder Testing*

Two tests were performed to identify optimal operating conditions for the rudder. The first was to verify the optimal stall angle for the rudder of 35°, taken from preexisting literature [6]. The second was to determine the effect if any, of the shape and size of the rudder on the turning ability of the rudder. The test setup, shown below in Figure 6, consists of a rudder bracket and propeller placed in a bucket of water. A spring is used to measure the force produced by the rudder.

For the first test, a single rudder is used and angles from 5° to 45° are tested. The results showed that the rudder angled at 35° produced the most force, after which the force significantly decreased.

The second test used only the 35° angle and compared the spring’s elongation with different rudders. Three rudder shapes were tested, each with three different chords such as to modify the surface area of each rudder (the span was maintained to ensure that all rudders would fit on the same bracket). Results of this test were inconclusive, and it is suggested to repeat this test to verify the results obtained.

More information can be found in Appendix F.

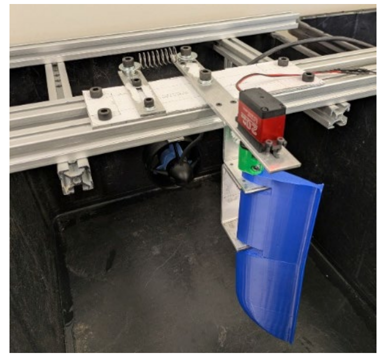

Fig. 6. Rudder Test Setup

B. *Electrical Testing*

First, isolated testing on the propellers was done directly off the RC remote. The purpose of this test was to verify the current draw of the propellers against the specifications and to confirm that they were below the trigger point of the breaker.

Testing to verify proper power cutoff for the remote e-stop and the on-board e-stop was done. Similarly, it was ensured that the breaker would trip at above 30 A.

To safely discharge the 14.8 V lithium battery, a BMS that monitors each of its cells was used and a voltage sensor was placed on its output. To test the functionality of the setup without needing to discharge the battery to an unsafe level, a variable power supply was connected, simulating a cell. The voltage level was set to 4.2 V, the maximum level of a cell, then brought down to 3.0 V, the minimum level of a cell. At this point, the voltage sensor stops having a reading, displaying 0 V on the RC remote.

C. *PID Testing*

A PID control algorithm [1] and testing environment were also developed to demonstrate the basic form of boat control. In this setup, an IMU was used to gather heading using its magnetometer, as this was originally how the boat’s heading would be gathered. A DC motor was used to turn the IMU until it was aligned with the target heading. The results of this test were a greater understanding on PID control, as well as tuning methods. It was determined that acquiring heading using the magnetometer is unreliable mainly due to the wires carrying fluctuating current through them. Various calibration methods for the magnetometer were tried with little success. The PID controller would be later used in the boat motor control node.

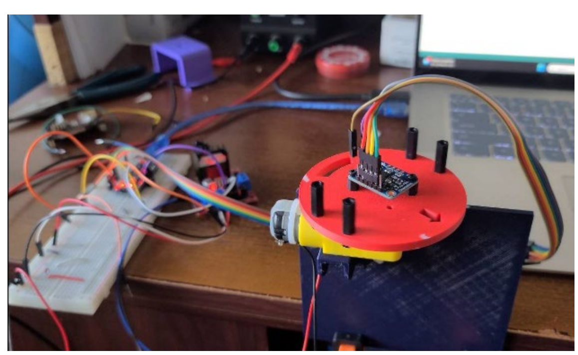

Fig. 7. PID and IMU test setup

D. GPS Testing

Various GPS devices were tested and compared, including the MT3337 Chip Set GPS module and the GPS units included in several Android smartphones (Motorola Moto G6, Samsung S20 Ultra, Google Pixel 7a). The objective of this test was to select the GPS which would be used in the Loon-E: Notable considerations included the accuracy, repeatability, and response time of the different GPS devices.

Testing was done in various conditions (ex. moving in a circle, moving in a straight line, standing still, etc.) to observe the longitude, latitude, speed over ground and track angle measurements provided by each device.

All GPS units used contained GNSS chips which did not use Real-Time-Kinematic GPS, and as such results were less accurate than a GPS with this functionality. From this testing, however, the Pixel 7a's GPS was found to be the most accurate (notably the most recently released of the phones tested).

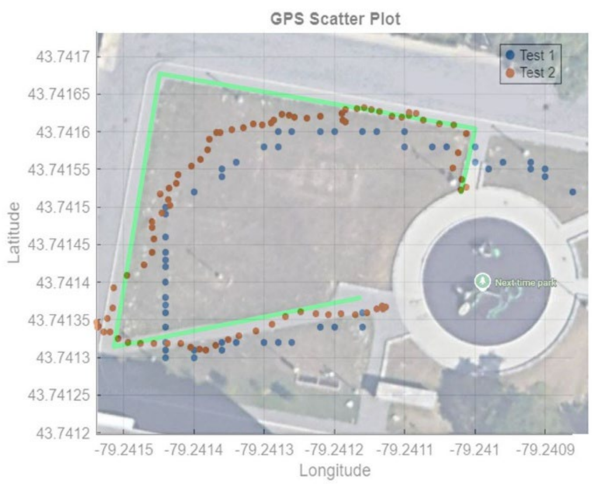

Fig. 8. Result of GPS coordinate test (Green: Actual Path, Blue: MT3337 Output, Red: Pixel 7a Output)

E. Object Detection Algorithm

3D models of course objects were recreated and printed, specifically those of the tall red and green buoys for the RoboBoat competition. By running a preliminary program which loads the model and runs processed images through it, the accuracy of the object detection algorithm can be verified. The model used during testing would need to be updated as the Njord challenge uses different objects from the RoboBoat competition; as such, the accuracy of this specific model was not as important as the verification that the overall methodology is effective.

Later, this was integrated into a preliminary version of the mapping and path planning algorithms to serve as a proof of concept and implemented in ROS 2. This was done with images from a single camera instead of the stereoscopic ZED X camera, and as such estimation of distance data was required. Errors encountered were a result of this estimation, but otherwise the methodology was shown to be effective. The example shown in Figure 9 shows the algorithm successfully identifying the space in the middle of the nearest pair of green and red buoys and making a path around the black buoy while leaving sufficient distance around the obstacle, as would be done for the RoboBoat competition.

0	0	0	0	0	0	0	0	0	3	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	3	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	3	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	3	0	0	0	0	0	0
0	0	0	0	0	7	0	0	0	3	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	3	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	3	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	3	0	0	0	0	0	0
0	0	0	0	0	0	0	3	3	0	0	0	0	0	0	0
0	0	0	0	0	0	0	3	0	0	0	0	0	0	0	0
0	1	1	0	0	0	0	3	0	0	0	0	0	2	2	0
0	1	1	0	0	0	0	0	0	0	0	0	0	2	2	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	1	1	0	0	0	0	0	0	0	0	0	0	2	2	0
0	1	1	0	0	0	0	0	0	0	0	0	0	2	2	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Fig. 9. Example Local Map with Obstacles (Red, Green, Grey) and Path Planning (Blue)

F. Simulation Environment

The team has limited budget and test space, and by extension has limited access to course objects such as buoys and dock cubes. This means that it is not possible to fully recreate the course environment in real life. To remedy this, a simulation environment was developed using Isaac Sim. This software was selected due to its compatibility with the selected hardware, namely the ZED X camera.

After building the simulation environment in Isaac Sim, Isaac Lab was used to pull the simulation file in and run some number of training simulations concurrently. This allowed the boat to make thousands of navigation attempts in parallel to accelerate the learning process. Each epoch of training runs 50-200 parallel models, selects the best fit model based on a reward function, then begins the next epoch by loading the best fit model with an additional 50-200 iterations. The program runs 15 epochs for each training session, results are reviewed, and the training algorithm is adjusted before re-running the training in Isaac Lab.

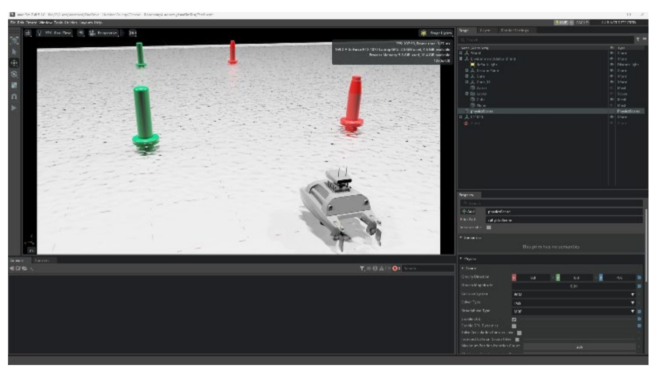

Fig. 10. Isaac Lab Simulation Environment

G. Real-World System Testing

To test the physical components of the boat, it was taken to a pond near Humber Polytechnic to verify the waterproofing of the system. This was done prior to the application of epoxy and consisted of placing the hull in the water for 10 minutes without any additional components. During this test, no water entered the interior of the boat where the electronics would be placed.

A second test was performed at a dock at Humber Bay, this time including any electrical components required to control propulsion. Key elements to be verified included the weight distribution and overall function of the electrical system. The boat was in the water for 15 minutes, during which it was observed that the boat was leaning back further than expected, even with the heaviest electrical components (ex. battery) being placed at the front of the boat. To account for this, any additional components should be placed at the front of the boat.

Throughout the year the electrical team made prototypes of the bigger system to ensure its functionality and performed testing in both dry and in-water environments.

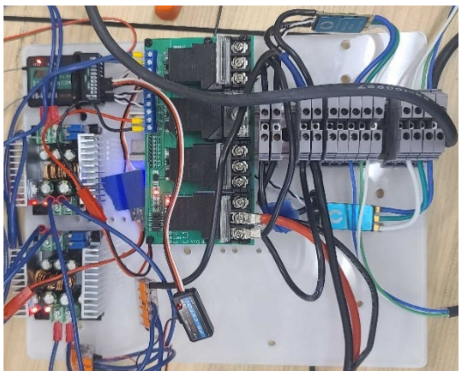

Fig. 11. Electrical components for RC driving test

H. Code Testing with Car

Testing of the programming prior to the setup of the simulation environment was also performed using a car. This car, the JetBot ROS AI Kit [7], was considered a fair approximation of the Loon-E as the motors controlling the JetBot's two wheels would operate under the same logic as the motors controlling the Loon-E's two propellers through differential thrust (though tuning of program parameters would be required when translating the code from one vehicle to the other). The JetBot was used to verify the code used in Task 1 and was able to effectively detect the 3D-printed buoys and navigate the path in both directions.

Fig. 12. Testing with JetBot

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[3] Roboflow, "Auto Label," *Roboflow Docs*, Jun. 23, 2025. https://docs.roboflow.com/annotate/ai-labeling/automated-annotation-with-autodistill

[4] G. Jocher, "Ultralytics YOLO11," *Ultralytics YOLO Docs*, Nov. 15, 2025. https://docs.ultralytics.com/

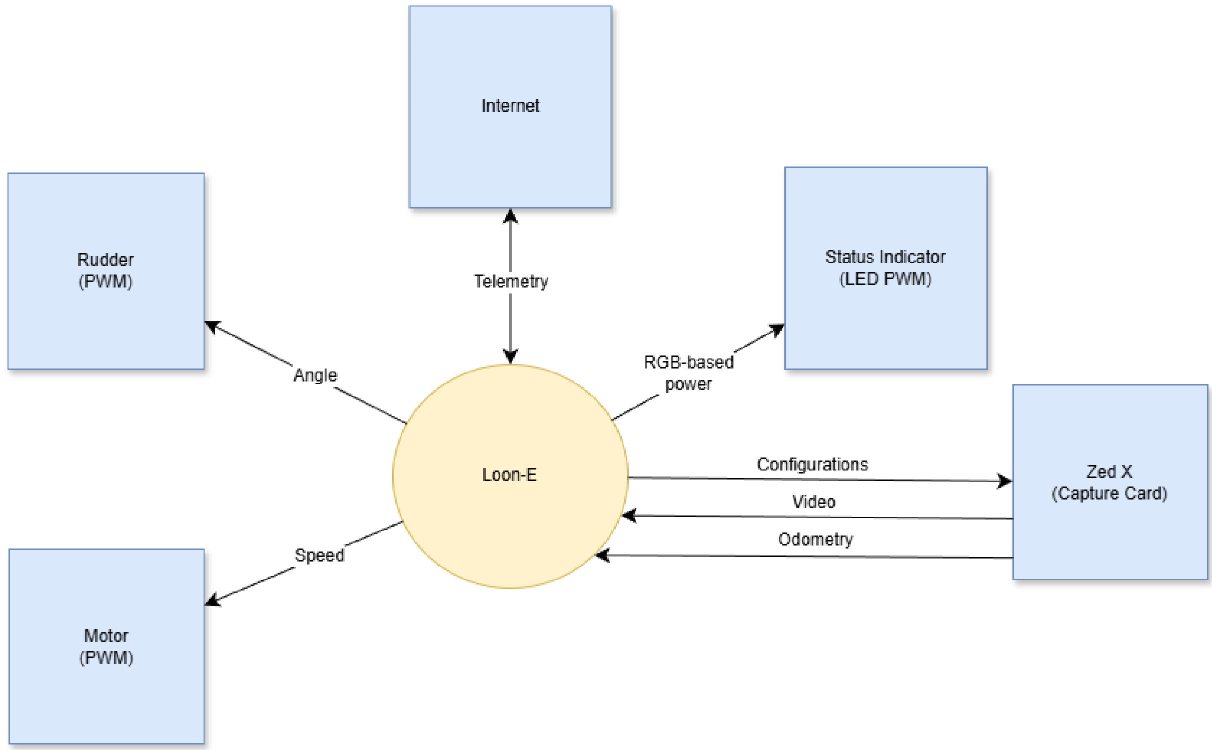
[5] P. Hart, N. Nilsson, and B. Raphael, "A Formal Basis for the Heuristic Determination of Minimum Cost Paths," *IEEE Transactions on Systems Science and Cybernetics*, vol. 4, no. 2, pp. 100–107, Jul. 1968, doi: https://doi.org/10.1109/tssc.1968.300136

[6] J. Liu and R. Hekkenberg, "Sixty years of research on ship rudders: effects of design choices on rudder performance," *Ships and Offshore Structures*, vol. 12, no. 4, pp. 495–512, May 2016, doi: https://doi.org/10.1080/17445302.2016.1178205

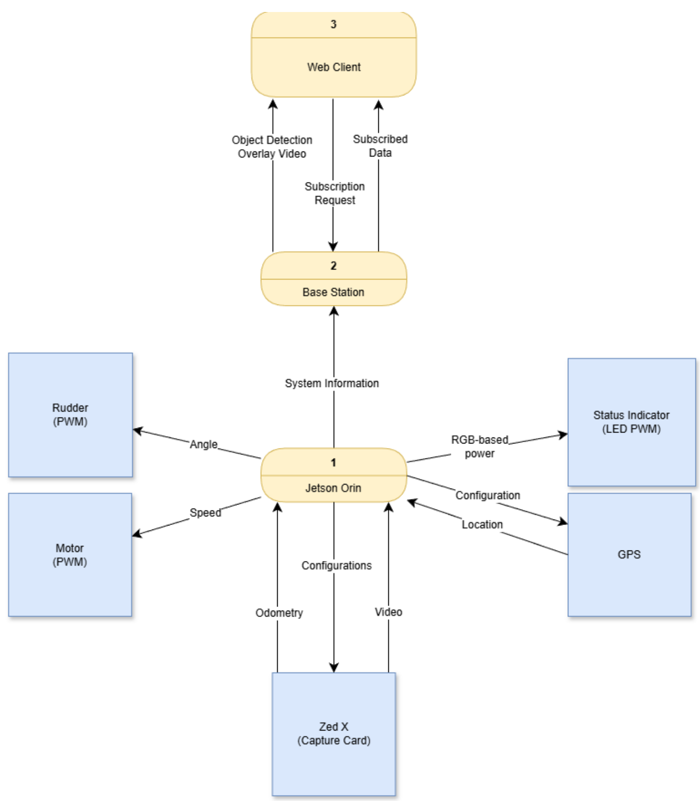
[7] Waveshare, "JetBot ROS AI Kit," *Waveshare Wiki*. https://www.waveshare.com/wiki/JetBot_ROS_AI_Kit

APPENDIX A: DATA FLOW DIAGRAMS

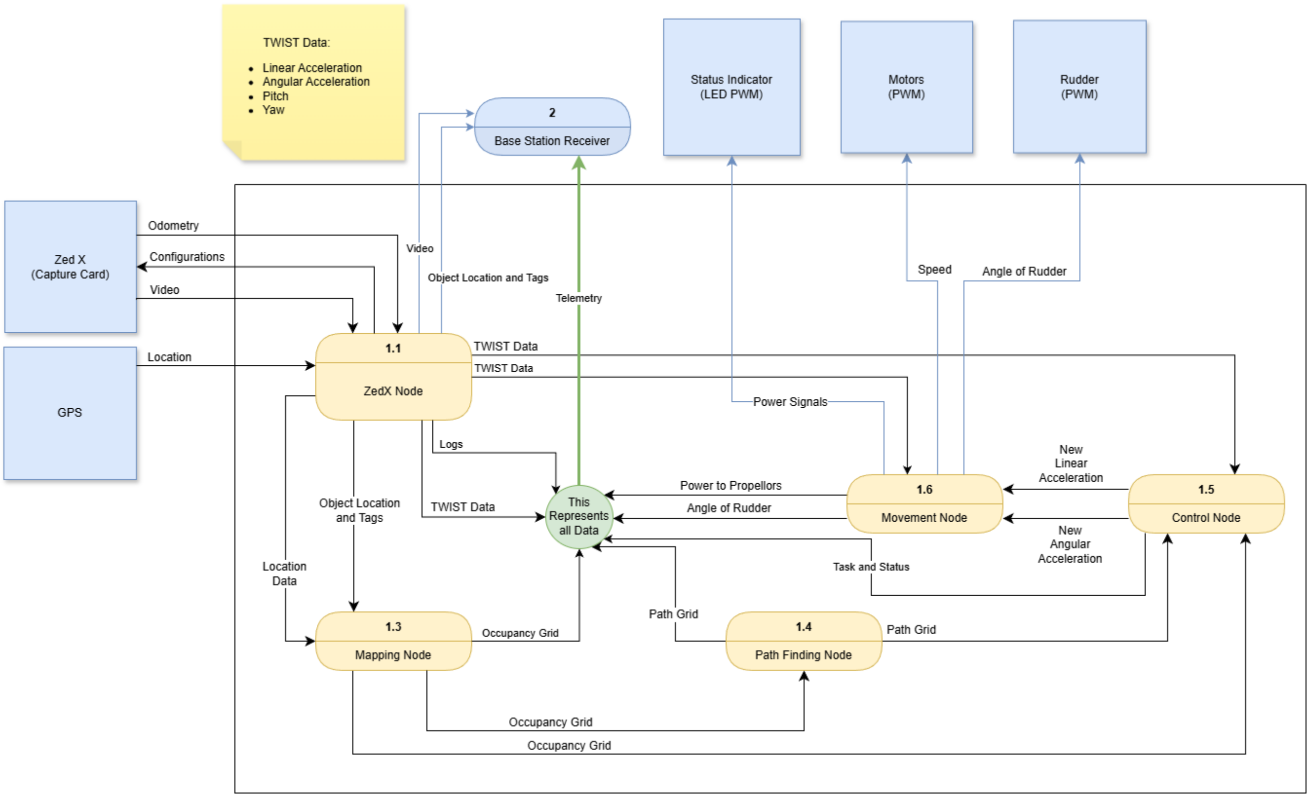
System Context Diagram



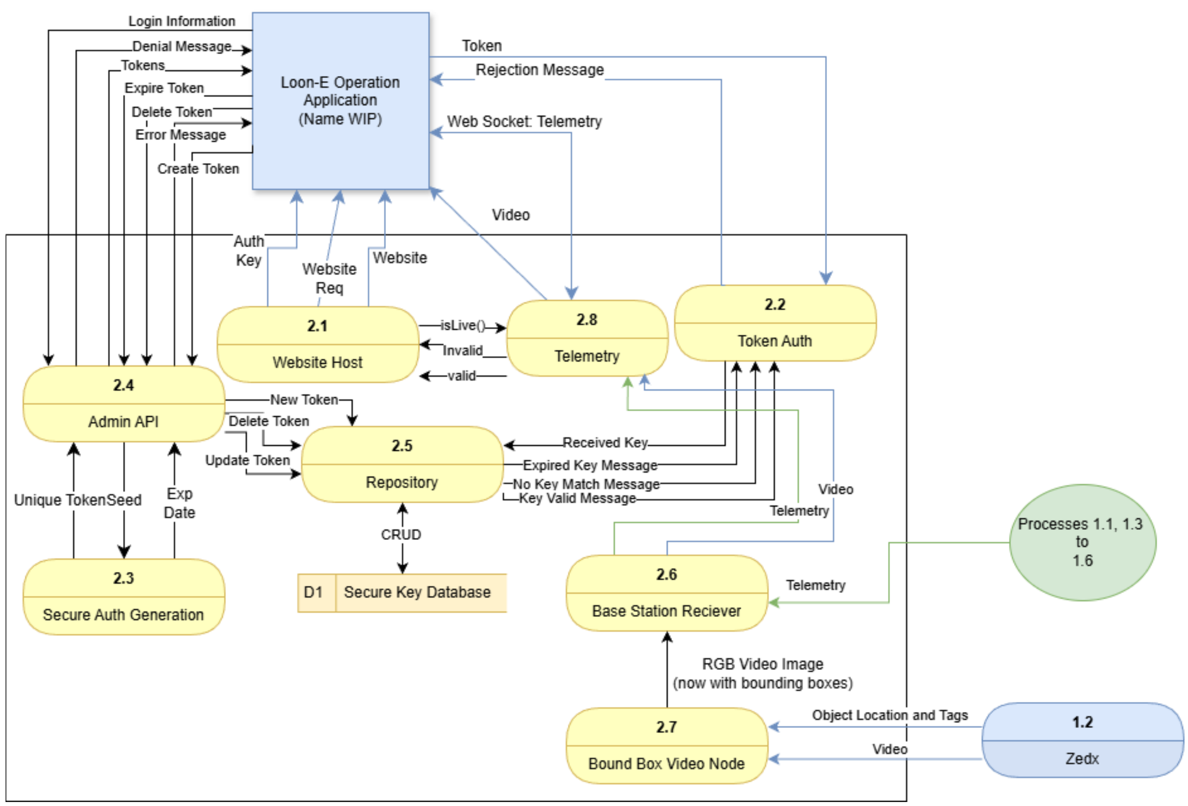
Level 1 Data Flow Diagram



Level 1.1 Data Flow Diagram for Primary Computer



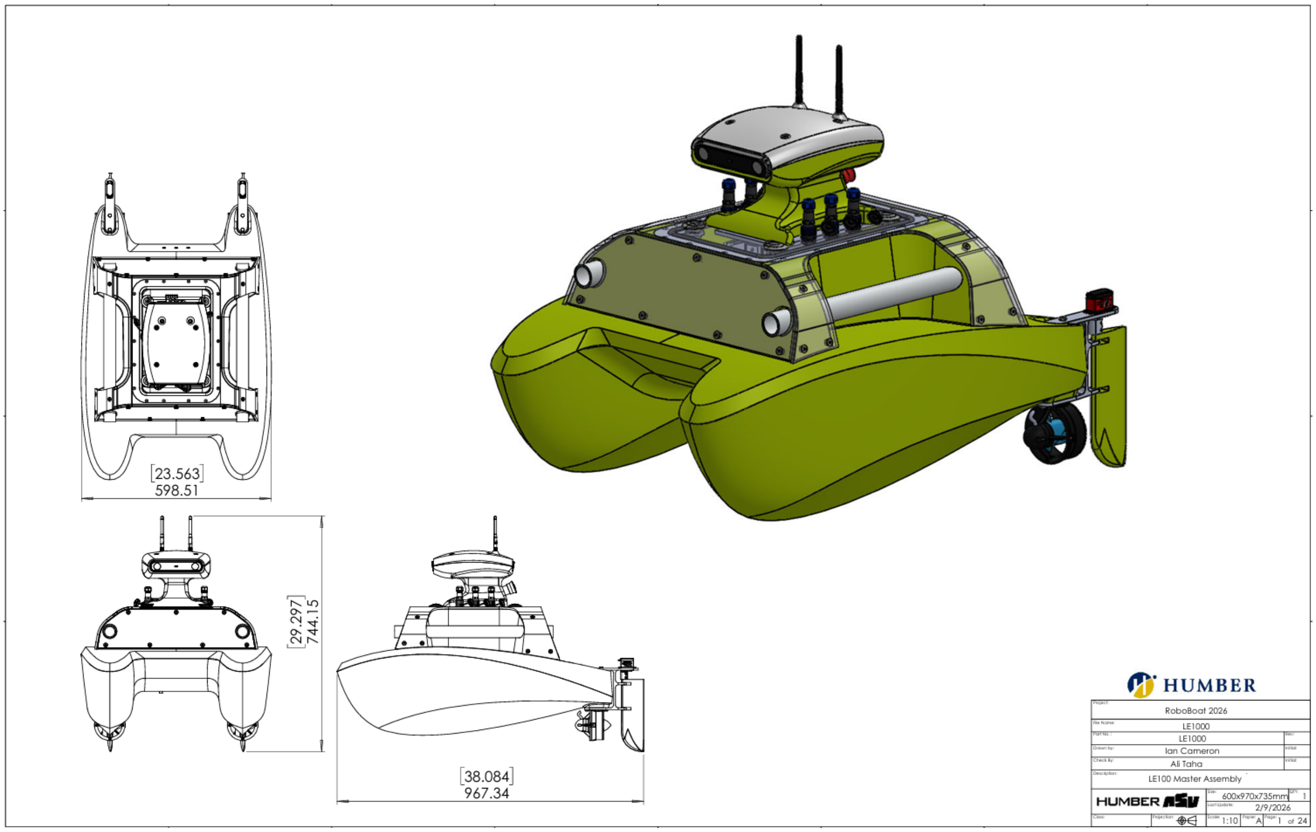
Base Station System Diagram

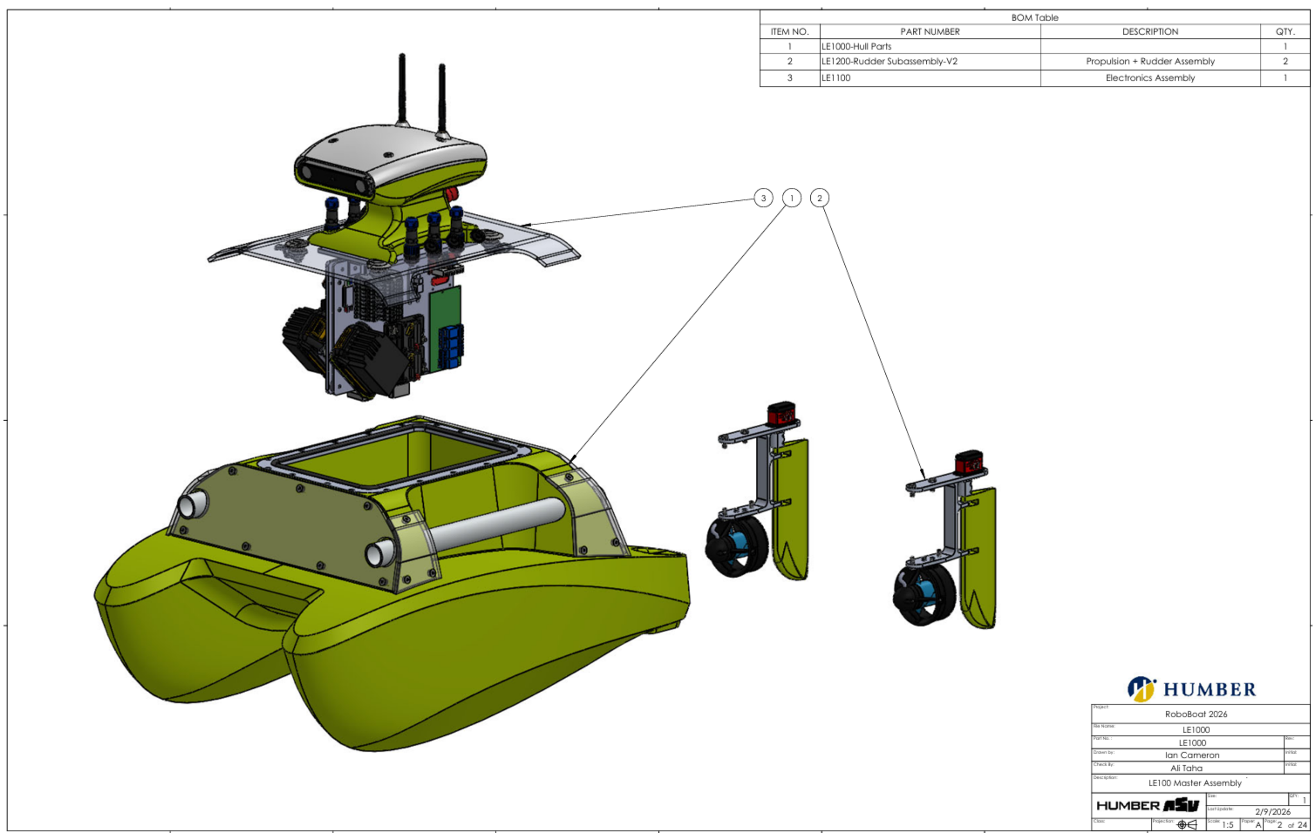


APPENDIX B: COMPONENTS LIST

SUBSYSTEM	COMPONENT	VENDOR	CHARACTERISTICS	COST/UNIT (USD)	QTY.
Propulsion	T200	Blue Robotics	31.21 A @ 20V	238	2
	Basic ESC	Blue Robotics	7-26 V	38	2
	Servo Motor	Miuzei	5V, 20 kg	15.32	2
Hull	Custom Hull	-	ABS	-	1
Remote Operation	FS-i6X	FlySky	10 channels	69.66	1
Navigation	Jetson Orin Nano	Nvidia	8GB	250	1
	ZED X Stereo Camera	ZED	Polarizer, 4mm	905	1
	ZED Link Capture Card	ZED	Duo	550	1
	Webcam	Logitech	1080p	30	1
	Pixel 7a	Google	Dual-Band GNSS	-	1
Electrical	Lithium-Ion Battery	Blue Robotics	14.8V, 18Ah	380	1
	20V Lithium-Ion Battery	DeWalt	20V, 6Ah	239	2
	Smart Dock	GoBILDA	20V	170	2
	Voltage sensor CVT01	Flysky	100V	17	1
	PCA9685	Adafruit	16 Channels	11.15	1
	Relay Module	YWBL-WH	30A	19.83	1
	HE Waterproof connector	HangTon	3, 4, 12 Pin	10	6
	Remote Control Electronic Switch	Fockety	3-30V, 20A	11.15	2
	E-Stop	McMasterCarr	2.5 A @ 24 V DC	52.44	1
	285-Series Circuit Breaker	Blue Sea Systems	30 A	75	2
	DC to DC voltage converter	Sayal Electronics	20A	18	3
	Custom Multiplexer PDB	-	RGB	20	1
	Custom Multiplexer PDB	-	16 input, 8 output	20	1
	Electronics enclosure	Hammond	ABS		1
Communications	EAP-225	TP Link	2.4 GHz + 5GHz	80	1
	XPS 15 9530	Dell		1800	1

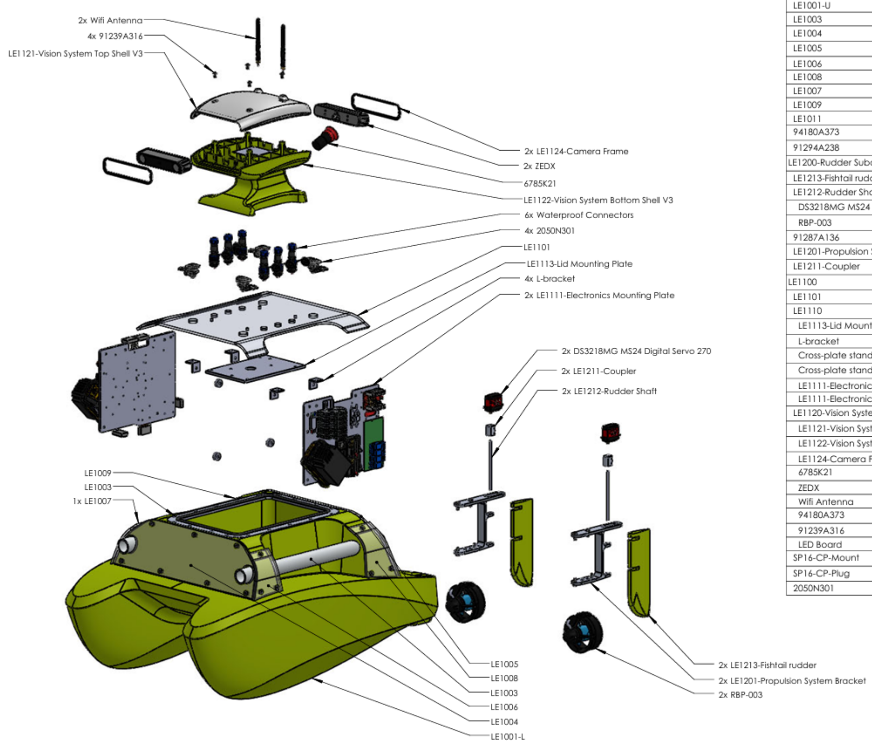
APPENDIX C: CAD DRAWINGS





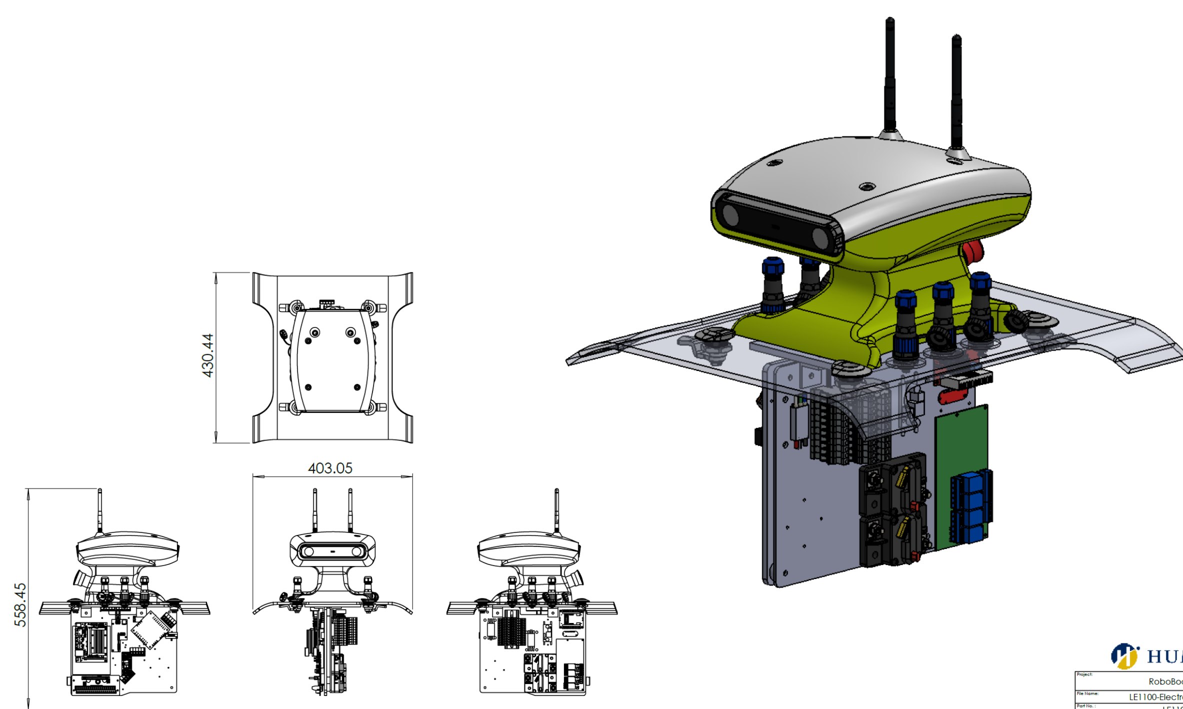
BOM Table

ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	LE1000-Hull Parts		1
2	LE1200-Rudder Subassembly-V2	Propulsion + Rudder Assembly	2
3	LE1100	Electronics Assembly	1

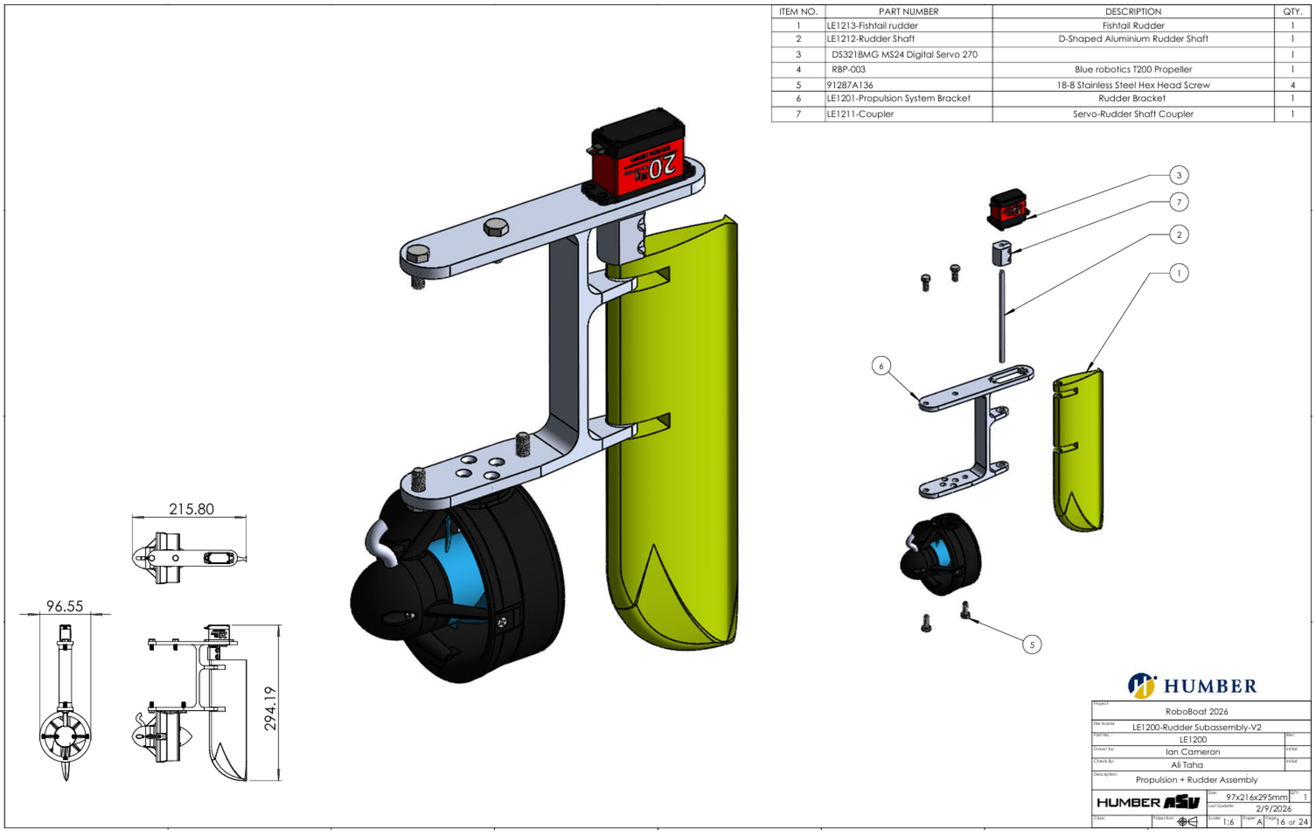


PART NUMBER	DESCRIPTION	QTY.
LE1000-Hull Parts		1
LE1001-L		1
LE1001-U		1
LE1003	Hatch Seal Frame	3
LE1004	Front Mounting Plate	1
LE1005	Back Acrylic Mounting Plate	1
LE1006		1
LE1008		1
LE1007		1
LE1009		1
LE1011		1
94180A373	Brass Tapered Heat-Set Inserts for Plastic	29
91294A238	Black-Oxide Alloy Steel Hex Drive Flat Head Screw	26
LE1200-Rudder Subassembly-V2	Propulsion + Rudder Assembly	2
LE1213-Fishtail rudder	Fishtail Rudder	1
LE1212-Rudder Shaft	D-Shaped Aluminium Rudder Shaft	1
DS3218MG MS24 Digital Servo 270		1
RBP-003	Blue robotics T200 Propeller	1
91287A136	18-8 Stainless Steel Hex Head Screw	4
LE1201-Propulsion System Bracket	Rudder Bracket	1
LE1211-Coupler	Servo-Rudder Shaft Coupler	1
LE1100	Electronics Assembly	1
LE1101	Electronics Hatch Lid	1
LE1110	Electronics Assembly	1
LE1113-Lid Mounting Plate	Electronics Mounting Plate	1
L-bracket		4
Cross-plate stand offs		3
Cross-plate stand offs_canal		1
LE1111-Electronics Plate 1-Power	Assembled Power Plate	1
LE1111-Electronics Plate 2-Electronics	Assembled Electronics Plate	1
LE1120-Vision System		1
LE1121-Vision System Top Shell V3	Plastic Top Shell LED Diffuser	1
LE1122-Vision System Bottom Shell V3	Lower Housing for Vision System	1
LE1124-Camera Frame		2
6785K21	Emergency Stop Enclosed Push-Button Switch	1
ZEDX		2
Wifi Antenna		2
94180A373	Brass Tapered Heat-Set Inserts for Plastic	8
91239A316	Button Head Hex Drive Screw	4
LED Board		1
SP16-CP-Mount	Waterproof Connector	6
SP16-CP-Plug	Waterproof Plug	6
2050N301	Hex-Drive Cam Lock	4

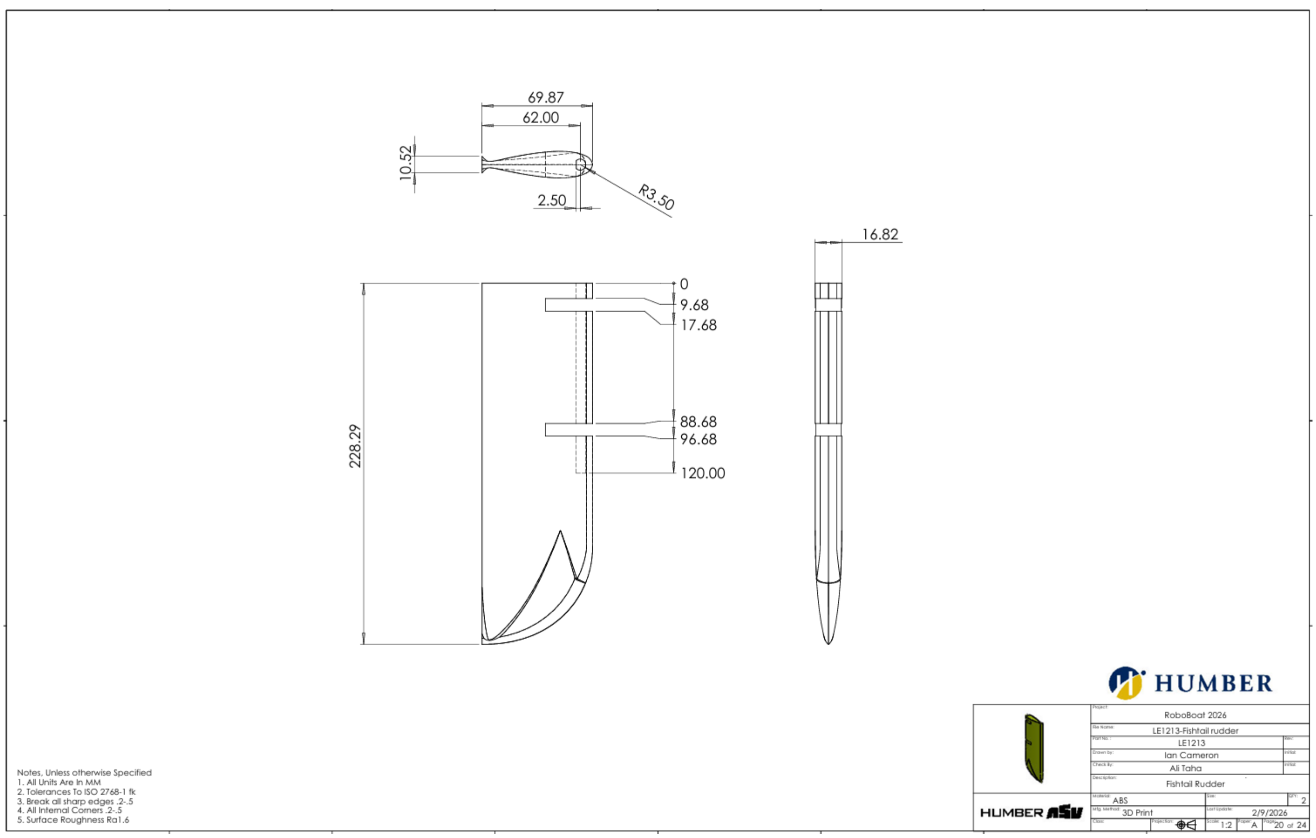
HUMBER
Project: RoboBoat 2026
File Name: LE1000
Part No.: LE1000
Drawn by: Ian Cameron
Check By: Ali Taha
Description: LE100 Master Assembly
HUMBER ASV
Last update: 2/9/2026
Qty: 1
Scale: 1:8 Paper: A Page: 3 of 24



HUMBER
Project: RoboBoat 2026
File Name: LE1100-Electronics Hatch
Part No.: LE1100
Drawn by: Ian Cameron
Check By: Ali Taha
Description: Electronics Assembly
HUMBER ASV
Size: 404x430x550mm
Qty: 1
Last Update: 01-Jun-2026
Scale: 1:8 Paper: A Page: 4 of 24



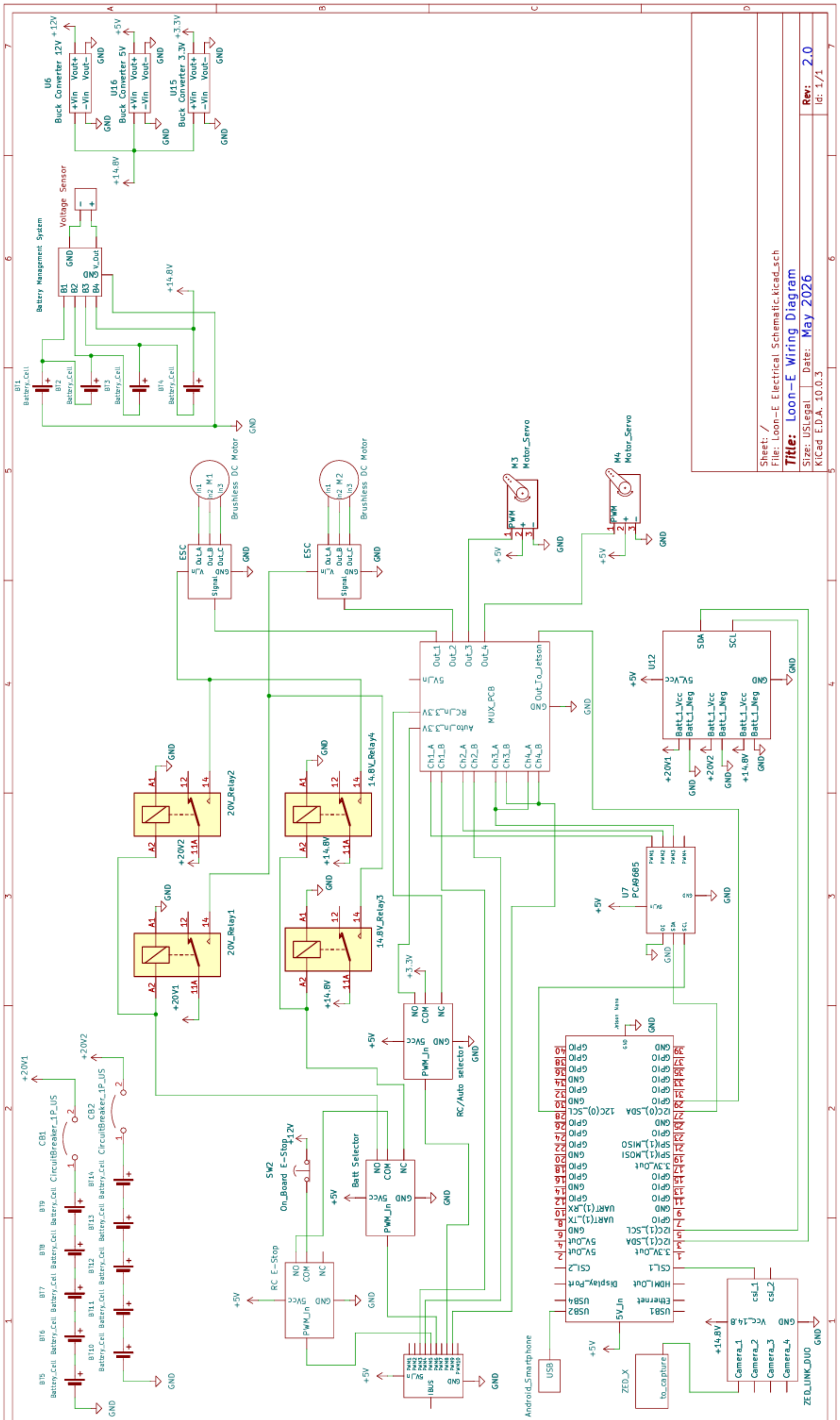
ITEM NO. | PART NUMBER | DESCRIPTION | QTY.
1 | LE1213-Fishtail rudder | Fishtail Rudder | 1
2 | LE1212-Rudder Shaft | D-Shaped Aluminium Rudder Shaft | 1
3 | DS3218MG MS24 Digital Servo 270 | | 1
4 | RBP-003 | Blue robotics T200 Propeller | 1
5 | 91287A136 | 18-8 Stainless Steel Hex Head Screw | 4
6 | LE1201-Propulsion System Bracket | Rudder Bracket | 1
7 | LE1211-Coupler | Servo-Rudder Shaft Coupler | 1
215.80
96.55
294.19
HUMBER
RoboBoat 2026
LE1200-Rudder Subassembly-V2
LE1200
Ian Cameron
Ali Taha
Propulsion + Rudder Assembly
97x216x295mm
2/9/2026
1:6
A
16 of 24
HUMBER ASV



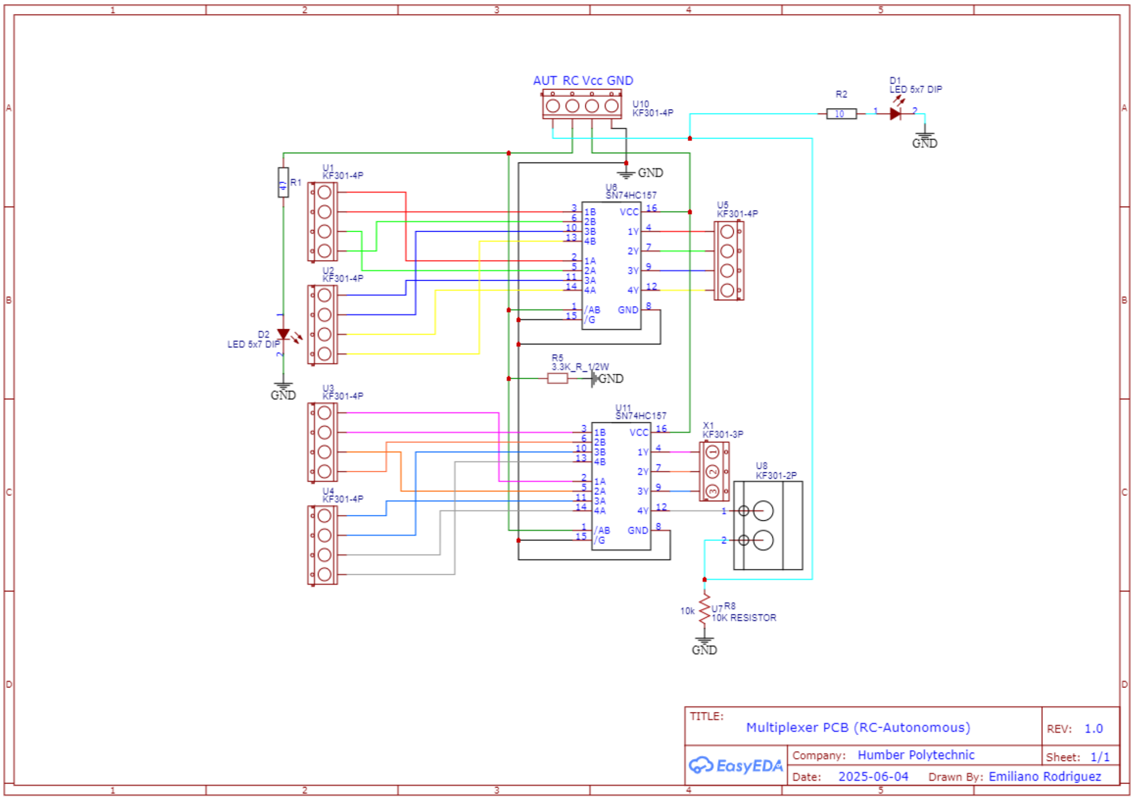
69.87
62.00
10.52
2.50
R3.50
16.82
0
9.68
17.68
88.68
96.68
120.00
228.29
HUMBER
RoboBoat 2026
LE1213-Fishtail rudder
LE1213
Ian Cameron
Ali Taha
Fishtail Rudder
ABS
3D Print
2/9/2026
1:2
A
2
20 of 24
HUMBER ASV
Notes, Unless otherwise Specified
1. All Units Are In MM
2. Tolerances To ISO 2768-1 fk
3. Break all sharp edges .2-.5
4. All Internal Corners .2-.5
5. Surface Roughness Ra1.6

APPENDIX D: ELECTRICAL DIAGRAM AND PCB

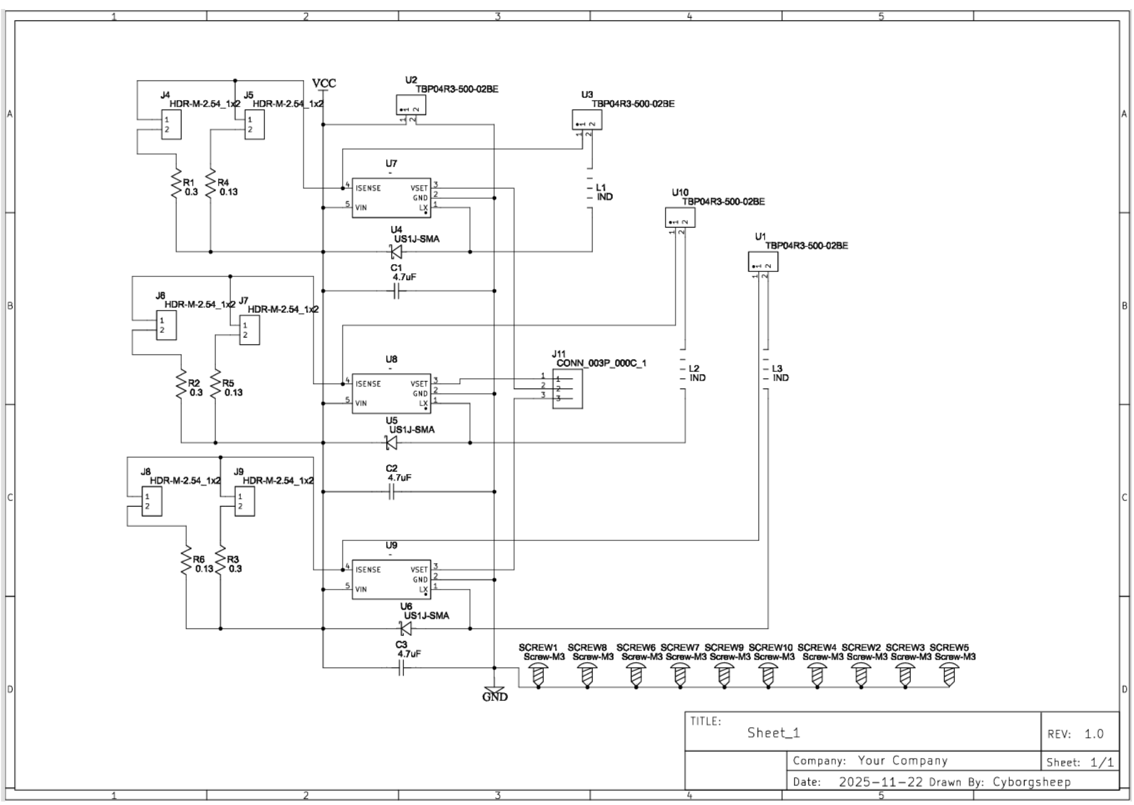
Loon-E Electrical Schematic 2026



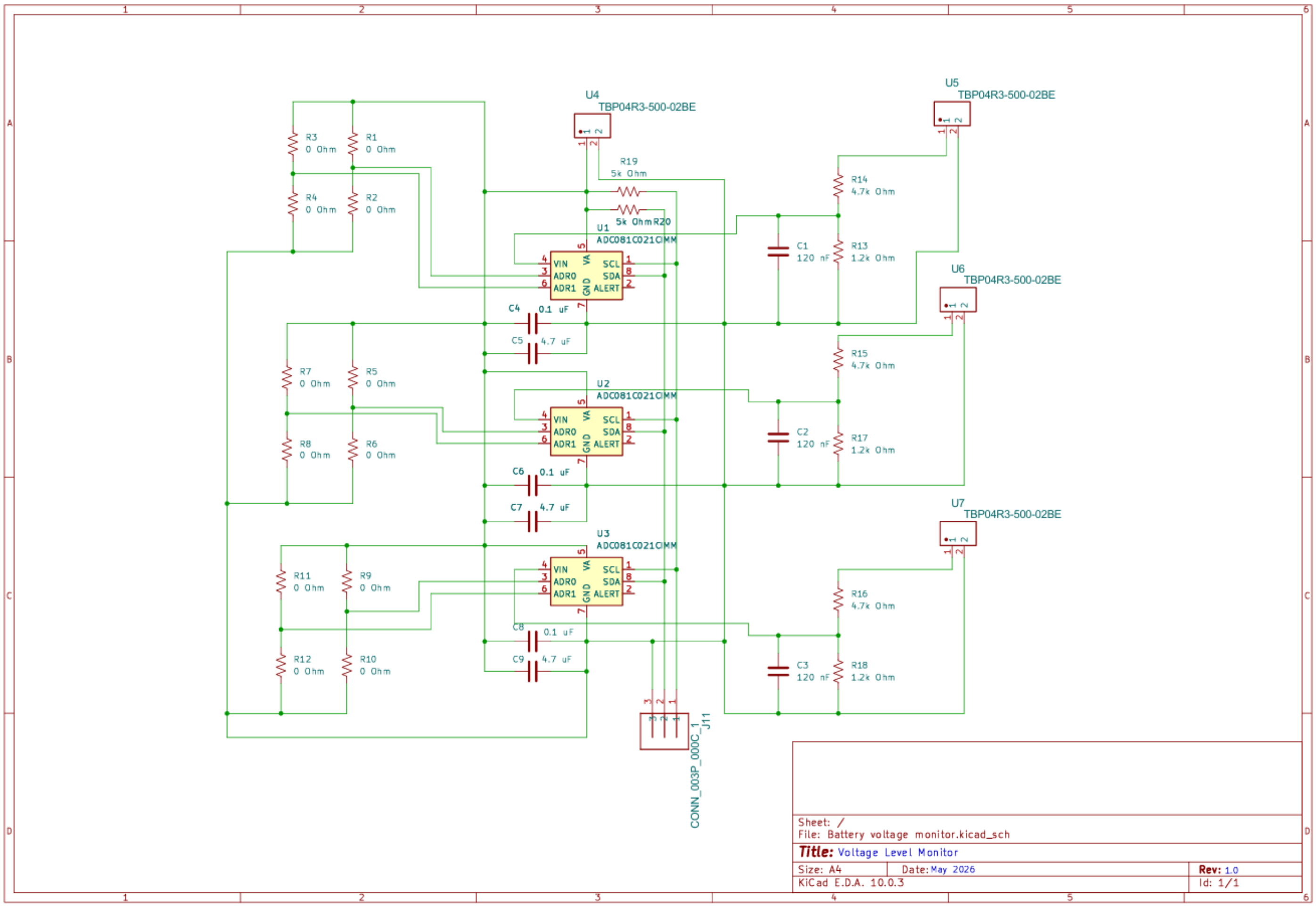
Multiplexer PCB



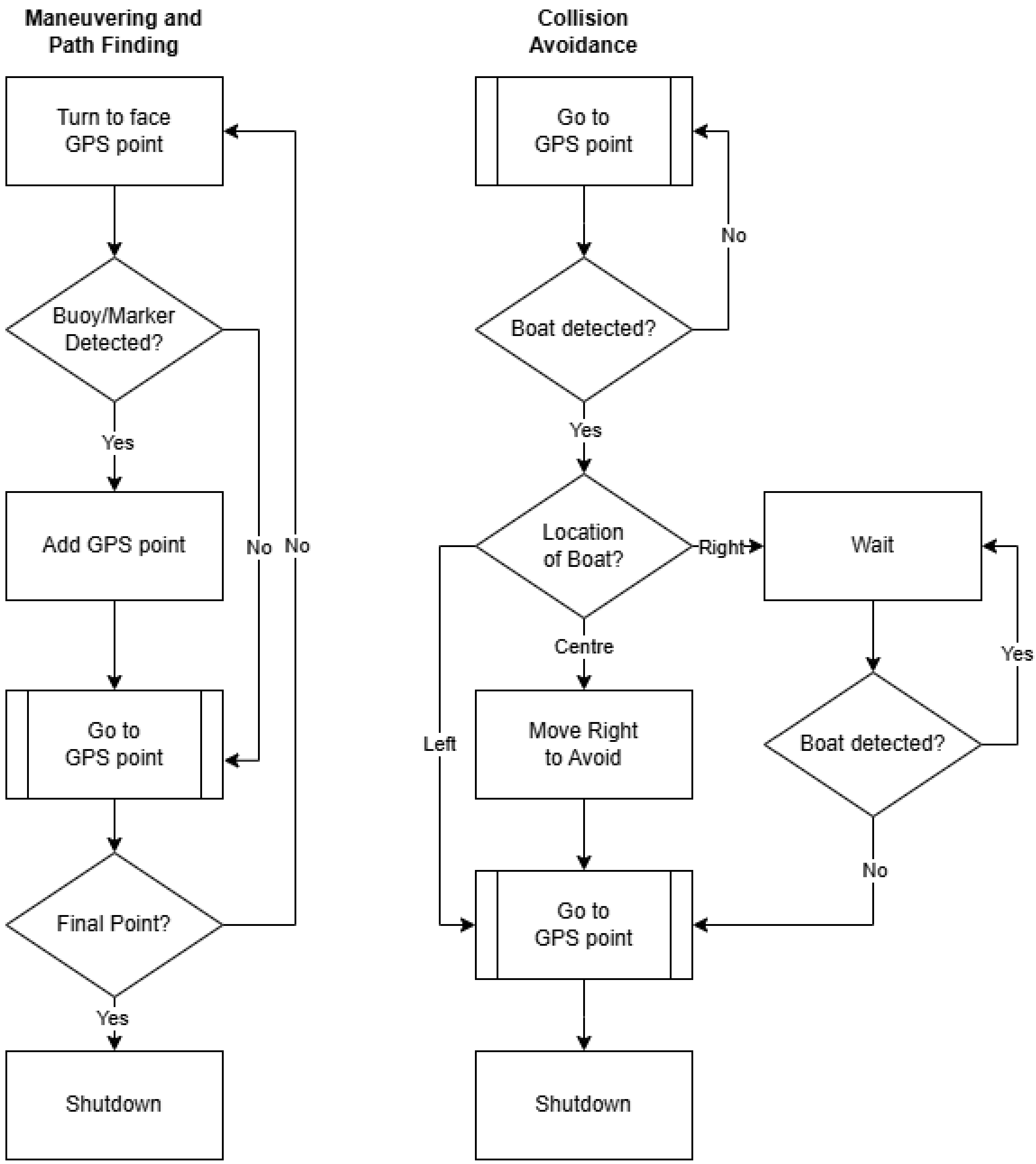
Light Indicator

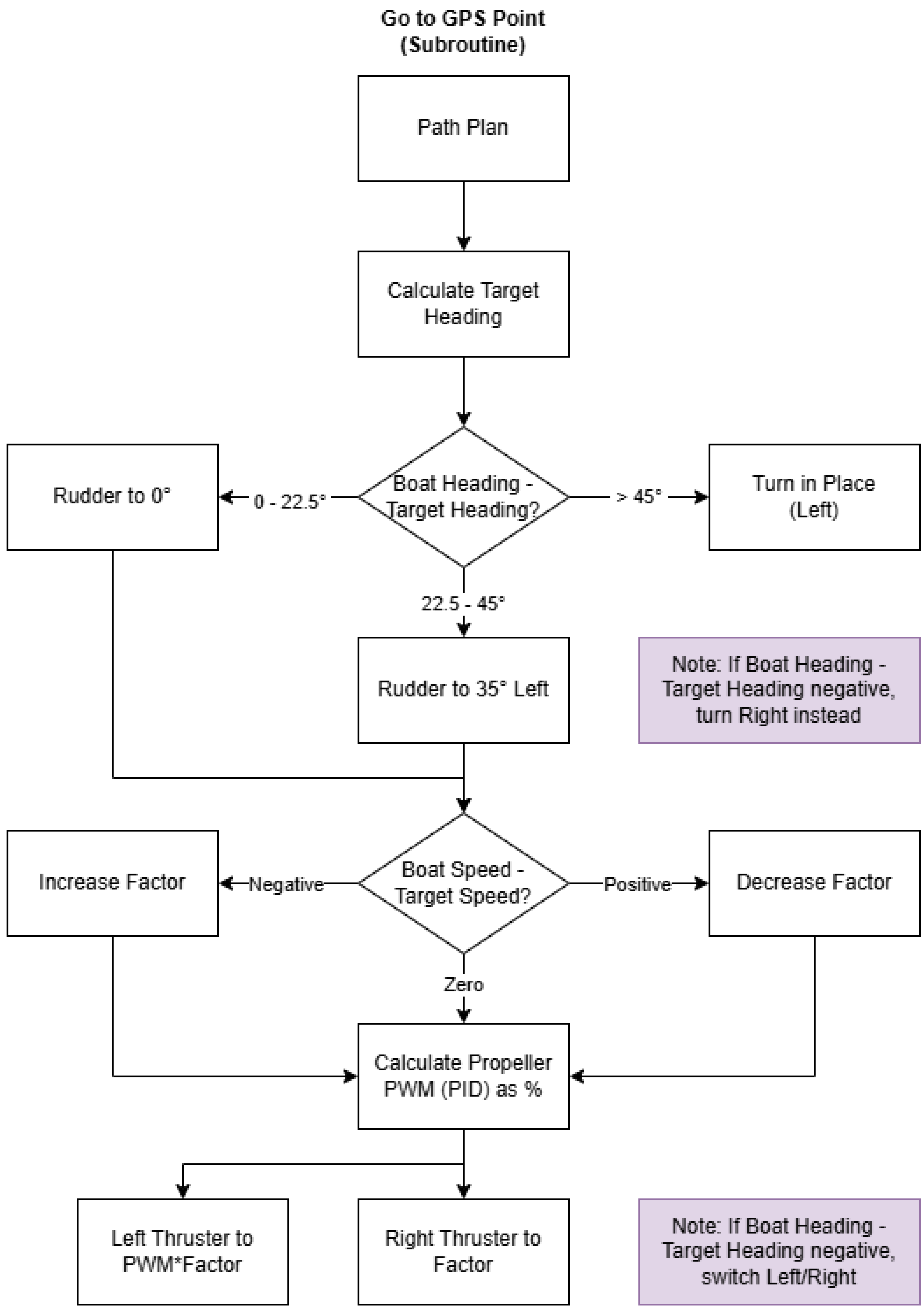


Voltage Sensor



APPENDIX E: TASK LOGIC FLOWCHARTS




Go to GPS Point
(Subroutine)
Path Plan
Calculate Target Heading
Boat Heading - Target Heading?
0 - 22.5°
Rudder to 0°
> 45°
Turn in Place (Left)
22.5 - 45°
Rudder to 35° Left
Note: If Boat Heading - Target Heading negative, turn Right instead
Boat Speed - Target Speed?
Negative
Increase Factor
Positive
Decrease Factor
Zero
Calculate Propeller PWM (PID) as %
Left Thruster to PWM*Factor
Right Thruster to Factor
Note: If Boat Heading - Target Heading negative, switch Left/Right

APPENDIX F: TEST RESULTS

Rudder Testing

Spring K Factor:

- Weight (m = 981 g) caused change in spring length (Δx) from 3.6 cm to 5.5 cm

$$mg = k\Delta x$$

$$k = \frac{0.981\ kg * 9.81\frac{m}{s^2}}{0.055\ m - 0.036\ m} = 506.5\ N/m$$

- Elongation was not completely in one axis, so 500 N/m is used as an approximation

Servo PWM values are assumed to be linear, with a 5° increase corresponding to a decrease of 40 μs. Three shapes of rudders were developed, each with three different sizes. Propeller current and thrust at different PWM values, as provided by Blue Robotics [27], are assumed to be accurate; A 16V battery is used for this test. Additional initial conditions of the testing are shown below:

Table 1a: PWM values

Angle (°)	PWM (μs)
0	1500
5	1460
15	1380
25	1300
35	1220
45	1140

Table 1b: Testing Conditions

Attribute	Value
PWM Propeller	1360 μs
Thrust Propeller	0.82 kg f
Rudder Span	226 mm
Rudder Chord S	52 mm
Rudder Chord M	70 mm
Rudder Chord L	87 mm

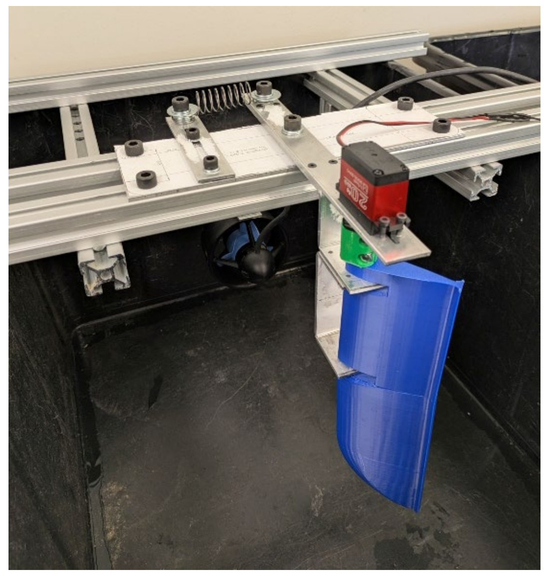
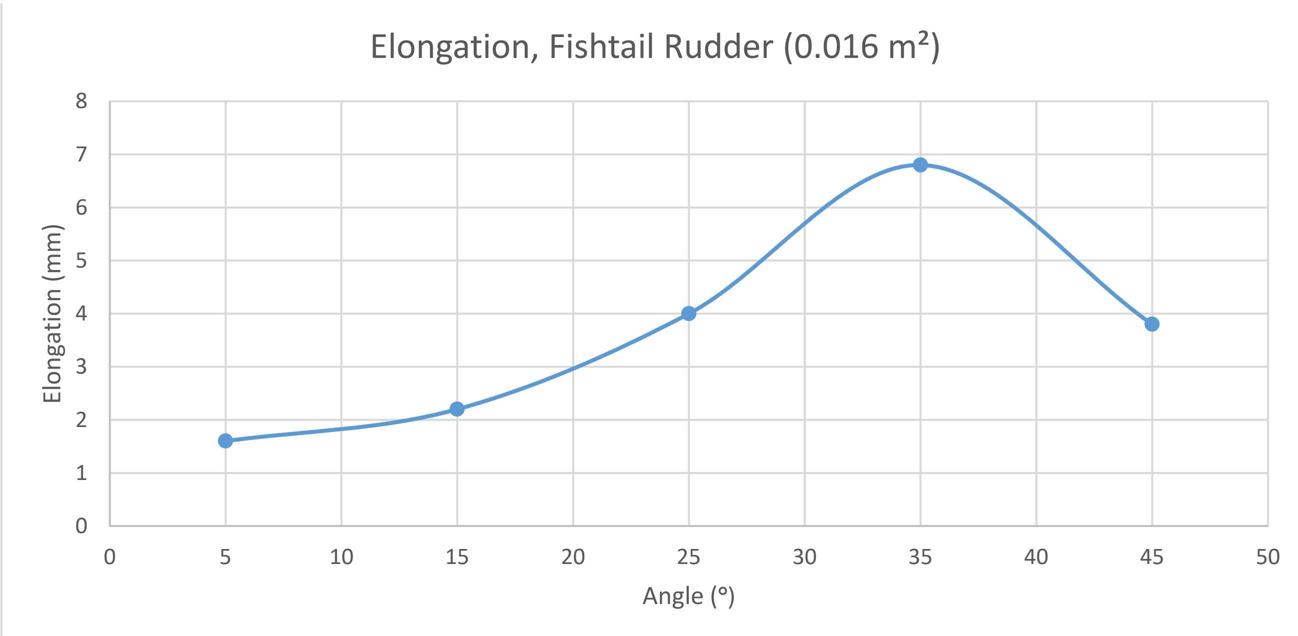


Figure 1a (left): Large Fishtail, Medium Triangle, and Small Oval Rudders (left to right)
Figure 1b (right): Test setup

Table 2: Results of Varying Rudder Angle

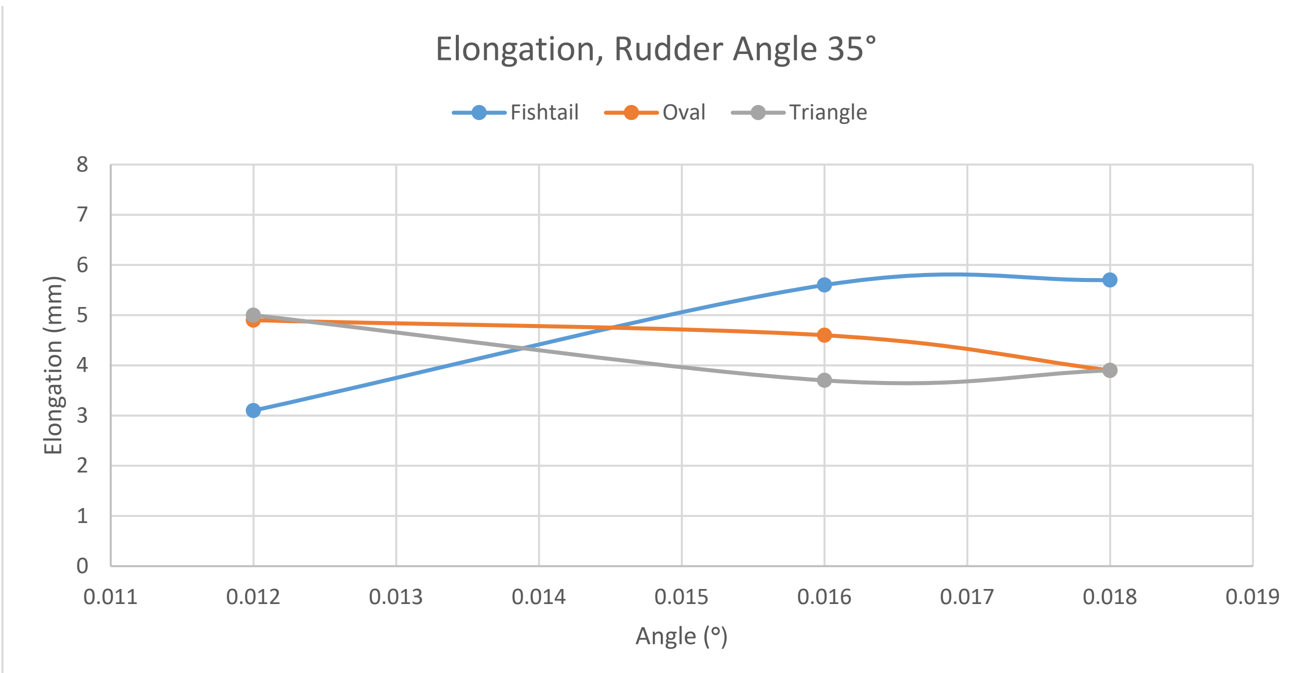
Rudder	Area (m^2)	Angle (°)	Initial (mm)	Final (mm)	Elongation (mm)	Force (N)
Fishtail	0.016	5	40.6	42.2	1.6	800
Fishtail	0.016	15	40.9	43.1	2.2	1100
Fishtail	0.016	25	40.8	44.8	4	2000
Fishtail	0.016	35	41.1	47.9	6.8	3400
Fishtail	0.016	45	40.4	44.2	3.8	1900



The results match the trend found in [29], though deriving the lift and drag coefficients from the force measured would not be possible with these values alone such as to more directly compare the two.

Table 3: Results of Varying Rudder Shape and Size

Rudder	Area (m^2)	Angle (°)	Initial (mm)	Final (mm)	Elongation (mm)	Force (N)
Fishtail	0.012	35	41.1	44.2	3.1	1550
Fishtail	0.016	35	40.4	46	5.6	2800
Fishtail	0.018	35	41.1	46.8	5.7	2850
Oval	0.012	35	40.5	45.4	4.9	2450
Oval	0.016	35	41	45.6	4.6	2300
Oval	0.018	35	41.4	45.3	3.9	1950
Triangle	0.012	35	40.5	45.5	5	2500
Triangle	0.016	35	41	44.7	3.7	1850
Triangle	0.018	35	40.7	44.6	3.9	1950



While results for optimal rudder angle successfully identify a stall angle of 35°, results for optimal rudder shape are less conclusive and in fact do not match with expected theoretical outcomes. The fishtail rudder matches with expected outcomes, while the oval rudder would match with expected outcomes only if 35° exceeds the stall angle (which would itself be unexpected). The triangle rudder's results do not seem to match with any expected outcome.

It is possible that the spring's high spring constant resulted in insignificant changes which were difficult to measure precisely using this setup, given that measurements were done by hand using a caliper; the difference between the largest and smallest measurements obtained is 2.6 mm. It is alternatively possible that the differences are insignificant between each rudder. However, even without this consideration, the forces produced assuming a correct spring factor are far too high, suggesting that the test setup itself is in some way flawed. Further testing, especially in-water testing, would be required to investigate these hypotheses.